

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		the pressure is higher than at sea level and the temperature is lower ☒ the pressure and temperature are both lower than at sea level ☐ particles are closer together at a higher pressure ☒ carbon dioxide is not a gas at any temperature or pressure ☐ particles are closer together at a higher temperature ☐ award (1) for each correct answer deduct (1) for each additional tick if more than two ticks			2	2		
		(ii)		award (1) each for any two sensible answers e.g. - proposal 2 makes use of the captured carbon dioxide (whereas it is only stored in proposal 1) – accept named uses - proposal 1 creates huge technical / engineering challenges - proposal 1 could lead to leakage of carbon dioxide (back to atmosphere / into the ocean) - proposal 1 creates plastic waste			2	2		
	(b)			$\text{CO}_2 + \text{NaOH} \rightarrow \text{NaHCO}_3$ award (1) for reactants award (1) for product deduct (1) if all formulae are correct but there has been an incorrect attempt to balance the equation		2		2		
				Question 5 total	0	2	4	6	0	0